California Real Estate IDX Exchange Reporting System

Technical Architecture & Pipeline Documentation

System Overview

This document describes a California real estate reporting system built on a Python-based data pipeline feeding into two Tableau dashboards. The system extracts monthly sold and listing data from the CoreLogic Trestle API, processes it into analytical datasets, and surfaces market trends, agent rankings, and office performance through interactive Tableau workbooks.

Pipeline Summary

Script / File	Output	Purpose
crmls_sold.py	CRMLSSold{YYYYMM}.csv	Monthly sold-data extractor
analysis_sold.py	priceratio.csv	Clean, derive market metrics
crmls_listed.py	CRMLSListing{YYYYMM}.csv	Monthly listing extractor
analysis_listed.py	newlistings.csv	Combine & filter listings
CA Market Analysis.twbx	(Tableau dashboard)	Market analytics by geography
Agent/Office Report.twbx	(Tableau dashboard)	Top agent & office rankings

How the System Fits Together

The Python scripts are the data pipeline; the Tableau files are the reporting layer. The Tableau workbooks do not perform raw API extraction themselves — they depend entirely on the CSVs produced by the Python scripts.

Data flow:

- crmls_sold.py → monthly sold CSVs
- analysis_sold.py → priceratio.csv
- crmls_listed.py → monthly listing CSVs
- analysis_listed.py → newlistings.csv
- Both Tableau workbooks read priceratio.csv and newlistings.csv

Python Scripts

crmls_sold.py — Monthly Sold-Data Extractor

This script is the raw monthly sold-data downloader. It calls the CoreLogic Trestle OData endpoint, first requesting a bearer token from a secure proxy endpoint at idxexchange.com. It queries only records where MlsStatus = 'Closed' and CloseDate falls within a specific target month (currently January 2026), then writes results to CRMLSSold202601.csv. Pagination is handled via @odata.nextLink, enabling retrieval of more than 1,000 records.

Fields extracted include:

- Pricing: OriginalListPrice, ListPrice, ClosePrice
- Property attributes: beds, baths, sqft, lot size, year built
- Location: city, postal code, address, latitude, longitude
- Agent/office: listing office, buyer office, agent names, agent MLS ID
- Other: garage spaces, stories, association fee, school fields

analysis_sold.py — Sold-Data Preparation & Metric Engineering

This is the main sold-data preparation and metric-engineering script. It loads monthly sold CSVs from January 2024 through January 2026, concatenates them into a single dataframe, and performs cleanup and modeling-prep steps before exporting priceratio.csv.

Key transformations:

- Drops non-reporting columns
- Converts CloseDate to datetime and filters to >= 2023-01-01
- Keeps only PropertyType == 'Residential'
- Creates derived fields: soldyear, soldmonth, yrmo, priceratio (ClosePrice / OriginalListPrice), pricesqft (ClosePrice / LivingArea)
- Computes monthly IQR-based outlier thresholds for priceratio and DaysOnMarket, replacing outliers with NaN rather than deleting rows

Note: Starting in August 2024, some monthly sold datasets were sourced from a flaskapi/raw folder rather than generated directly from crmls_sold.py, indicating the workflow has evolved over time.

crmls_listed.py — Monthly New-Listing Extractor

The listing-side counterpart to crmls_sold.py. It uses the same secure token proxy and Trestle Property endpoint, but filters by ListingContractDate within the target month. Results are written to CRMLSListing202601.csv. Field selection and pagination logic are structurally identical to the sold extractor.

analysis_listed.py — Listing-Side Analytical Table Builder

A simpler counterpart to analysis_sold.py. It loads monthly listing CSVs from January 2024 through January 2026, concatenates them, filters to PropertyType == 'Residential', and exports newlistings.csv. Compared with analysis_sold.py, it performs significantly less transformation:

- No outlier handling
- No derived market ratios
- No date filtering beyond whichever months were loaded
- No major column dropping

The result is a straightforward combined archive of residential listings for Tableau consumption.

Tableau Workbooks

CA Market Analysis May 2025.twbx

A geographic market-reporting dashboard for California, built on priceratio.csv and newlistings.csv. Users can analyze market conditions by city, county, and ZIP code, and pull comparable sales views at the ZIP level.

Dashboards included:

- City, CityYOY
- County, CountyYOY
- Zipcode, ZipcodeYOY
- Homes Sold by Zip Code
- Median Price By Zip Code
- Zipcode Comps (list and map)

KPIs tracked:

- Median sales price
- Days on market
- Price per square foot
- Sold/list price ratio
- Homes sold year over year
- Listings and prices year over year

The workbook includes a Selected City parameter (defaulted to "29 Palms") for filtering views to a specific market.

California Listing Agent and Office Report — Top 100 Premium Edition.twbx

A premium ranking and reporting product for identifying top listing agents and offices by units and sales volume. Uses the same underlying CSVs (priceratio.csv and newlistings.csv) but focuses on competitive analysis and prospecting.

Dashboards included:

- Top Listing Agent — List
- Top Listing Agent — Map
- Top Listing Agents (Units) and (Volume)
- Top Listing Office (Units) and (Volume)
- Top Listing Office — List and Map

Parameters and key calculations:

- Selected City, Select Company 1/2, Select Agent 1/2, Select Top 50 Agent
- Units Sold Sort = COUNT([ClosePrice])
- Share of Homes Sold = COUNT([ClosePrice]) / SUM({ FIXED [PostalCode]: COUNT([ClosePrice]) })
- Agent/company comparison fields and top-50-agent filter

Key Difference Between the Two Workbooks

CA Market Analysis focuses on the market — answering questions like: What is median price doing? How are listings and sold counts changing? What is DOM or price per sqft in this area?

The Agent/Office Report focuses on people and brokerages — answering questions like: Who are the top listing agents? Which offices dominate a ZIP code? What share of homes sold belongs to specific agents or offices?

Summary

These files together form a complete California real estate reporting system:

- Extract sold and listing data monthly via the CoreLogic Trestle API
- Standardize, combine, and clean the raw data
- Derive market metrics (price ratio, price per sqft, DOM outlier handling) for sold data
- Feed two polished Tableau dashboards: one for geographic market analysis, one for agent and office rankings

The structure is logical and easy to follow. The primary operational burden is the monthly manual refresh of the analysis scripts as new CSV files are added.